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ISM 6561 Deep Learning
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FINAL PROJECT REPORT

AI-Driven Demand Forecasting and Dynamic Pricing Optimization in Omnichannel Retail

1. Introduction and Problem Specification

Targeted Industry and Importance

- I focus on the retail industry, specifically omnichannel retailing, where businesses operate both physical stores and e-commerce platforms. This industry is crucial because it drives global consumer spending, business growth, and supply chain activity.
- Retailers constantly face fluctuating demand patterns and competitive pressures. Demand forecasting and pricing strategy are essential to:
- Ensure product availability, prevent stockouts, and maintain healthy profit margins.

Why the Industry Is Important

Demand forecasting helps retailers better anticipate product demand, reduce stock-related inefficiencies, and improve planning decisions. Dynamic pricing helps retailers explore how pricing decisions influence demand and revenue over time. Together, forecasting and pricing optimization support more informed and adaptive retail decision-making.

The Importance of AI-Driven Forecasting & Pricing

- **Support Demand Forecasting**
  Deep learning models such as TFT provide a structured way to model multi-step retail demand and capture temporal patterns in sales behavior.
- **Explore Dynamic Pricing**
  A PPO-based pricing agent can simulate adaptive pricing strategies and study how price changes affect demand and cumulative reward over time.
- **Build an Interpretable AI Pipeline**
  Combining forecasting, pricing simulation, and explainability methods creates a strong foundation for scalable retail analytics.

2. Problem Statement

Retailers face two main challenges:
- Accurately forecasting demand across multiple SKUs and locations.
- Dynamically optimizing pricing in real time to maintain profitability.

Traditional forecasting and pricing models often fail to handle complex, multi-level data and fast-changing market conditions, leading to stockouts, overstocking, and lost revenue. My project addresses this using AI-driven methods (TFT + PPO) to build a practical forecasting and pricing optimization pipeline for retail demand analysis.

3. Research Questions: Driving Smarter Retail with AI-Based Forecasting and Pricing Optimization

These two core business questions shaped my project, guiding us to build a deep learning solution that enhances demand forecasting and dynamic pricing for retail optimization.

Question 1: How can deep learning models accurately forecast multi-step product demand using sales, calendar events, and pricing data to enable smarter inventory management and reduce stock-related risks?

This question explores how advanced forecasting can help retailers anticipate future demand with precision. Traditional forecasting methods often fail to capture complex patterns influenced by seasonality, events, and pricing, resulting in stockouts, overstock, and missed sales.

My project addresses this by:

- **Building a Forecasting Pipeline:** Leveraging Temporal Fusion Transformer (TFT) to model multi-step demand patterns using sales, calendar, and pricing features.
- **Improving Inventory Efficiency:** Ensuring optimal stock levels to meet customer demand while minimizing excess.
- **Understanding Demand Drivers:** Identifying how factors like holidays and price shifts impact demand to enable smarter planning.

Improved forecasting can support better inventory planning and operational decision-making by helping retailers anticipate demand more effectively.

Question 2: How can reinforcement learning dynamically optimize pricing strategies in real time to maximize revenue and profitability in a competitive retail environment?

This question targets building adaptive pricing strategies that automatically adjust prices in response to real-time market signals. Traditional pricing strategies often lag fast-changing market conditions, resulting in lost profit opportunities.

I aim to solve this by:

- **Pricing Optimization via Reinforcement Learning:** Using Proximal Policy Optimization (PPO) to learn adaptive pricing strategies in a simulated retail environment.
- **Enhancing Responsiveness:** Enabling real-time reactions to change in demand and competition.
- **Automating Decisions:** Allowing AI to explore and implement pricing strategies without manual intervention.
- AI-driven dynamic pricing provides a useful framework for studying how pricing decisions influence revenue and demand in changing retail conditions.

4. Dataset Overview and Data Characteristics

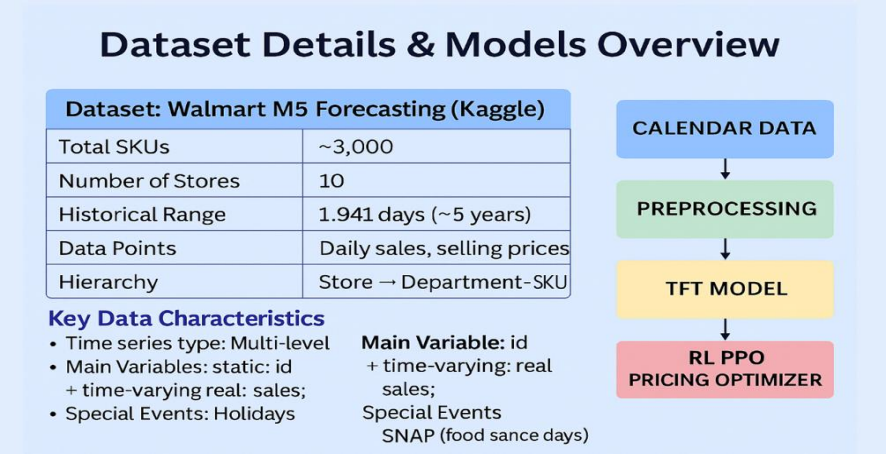
4.1 Dataset: Walmart M5 Forecasting (Kaggle)

This dataset includes detailed retail sales data from Walmart, covering approximately 3,000 products across 10 stores over a period of about 5 years. It provides a rich mix of sales, pricing, and calendar event data.

Main data files:

- Sales data: Daily unit sales for each product.
- Calendar data: Dates, holidays, SNAP (food assistance) events.
- Price data: Selling prices for each product and store.

Dataset Link: https://kaggle.com/competitions/m5-forecasting-accuracy



4.2 Data Characteristics

- Multi-level time series with sales tracked for each product-store combination.
- Static categorical features include item ID, store ID, department, category, and state.
- Time-varying features include calendar-based variables (date, weekday, month, year, holidays) and pricing data.
- The target variable is sales, with additional engineered features such as lagged sales, rolling averages, and calendar-based variables to capture short-term demand behavior and seasonality.

4.3 Data Preparation

I prepared the data in two main steps:

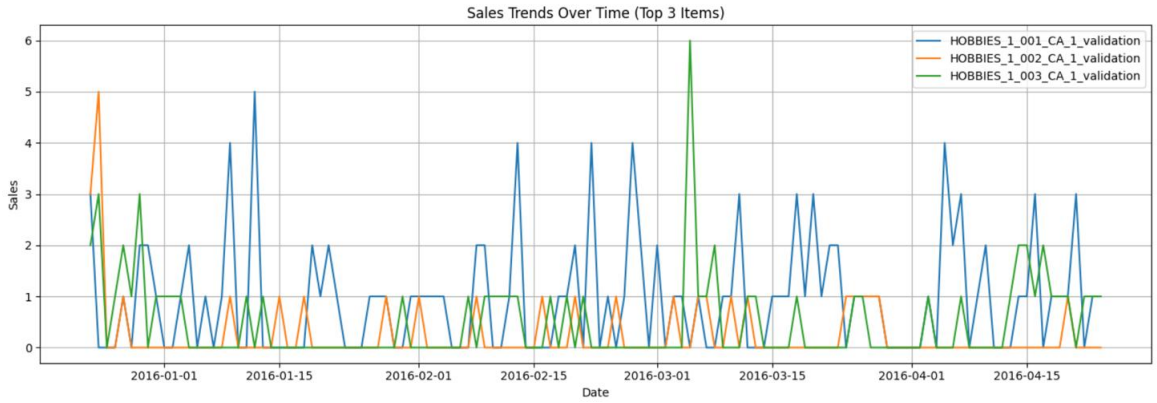
- **Filtering & Merging:** I focused on a single store (CA_1), selected the top 30 items, and restricted the analysis to the most recent 180 days to keep the problem computationally manageable. The sales data was reshaped from wide format to long format and then merged with calendar and selling price data to create a unified retail time-series dataset.
- **Feature Engineering:** The date column was converted to datetime format and used to generate a time index for forecasting. I created multiple lag-based features (lag_1, lag_7, lag_14) and rolling mean features (rolling_mean_7, rolling_mean_14) to capture

recent demand behavior and local trends. Additional calendar-based variables were engineered, including day_of_week, day_of_month, week_of_year, month, year, and is_weekend. Missing price values were forward-filled and backward-filled within each item-store group, and rows with missing values in required forecasting inputs were removed before constructing the TFT dataset. Categorical fields such as item, store, department, category, and state identifiers were converted into model-compatible formats.

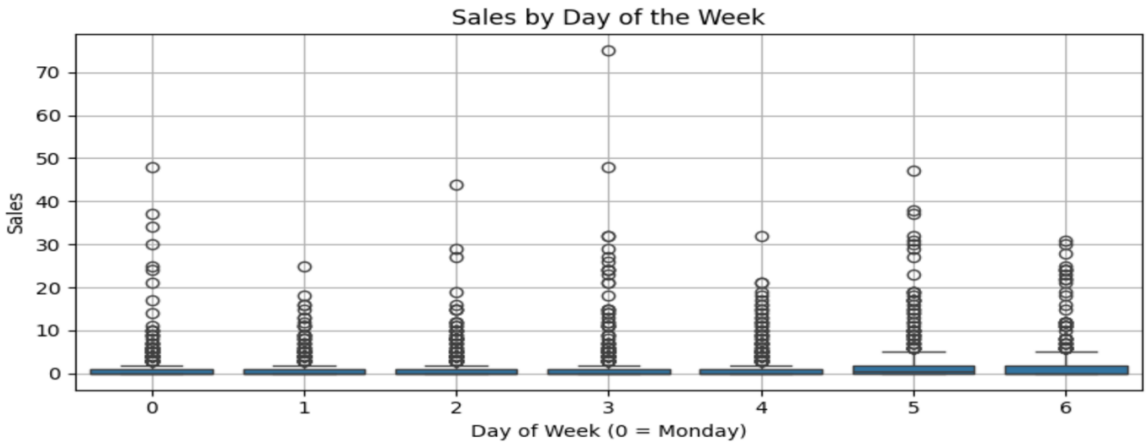
4.4 Exploratory Data Analysis (EDA) Visuals and Insights:

I explored sales patterns to understand demand trends and seasonality.
The EDA showed that sales are highly sparse and uneven over time, with visible short-term spikes and day-level variation. These observations motivated the use of lag features, rolling averages, and calendar-derived variables in the forecasting pipeline.

Sales Trends Over Time (Top 3 Items):
This line chart shows daily sales trends for three selected items over 180 days. It highlights sales spikes and seasonal fluctuations, confirming the need for time-based features in the model.



Sales by Day of the Week:
Sales by Day of the Week: This boxplot shows that sales vary across days of the week, with visible differences in spread and occasional spikes on certain days. This supports the inclusion of weekday-based features to capture weekly seasonality.



5. Methodology

5.1 Temporal Fusion Transformer (TFT) - Demand Forecasting

Model Definition (What it is & why I used it):
The TFT is a state-of-the-art deep learning model designed for **multi-step, multi-horizon time series forecasting.** It combines attention mechanisms and recurrent layers to capture both **short- and long-term patterns** in the data. I chose TFT because it can handle **mixed types of features (categorical + continuous)** and provide **interpretable results,** which is critical for business decision-making.

- **Why it's suitable:**
  It can model complex demand fluctuations driven by time (calendar effects), price, and product metadata—exactly matching my project's requirements.

Training Process (What I did):
I trained the TFT using the PyTorch Forecasting framework with a 7-day prediction horizon. The final configuration used a learning rate of 0.03, hidden size of 16, attention head size of 1, dropout of 0.1, and a maximum of 30 training epochs. The model used both static categorical features and time-varying engineered features, including lagged demand, rolling averages, and calendar-based variables such as day_of_week, day_of_month, week_of_year, month, year, and is_weekend. Early stopping and checkpointing were applied during training to retain the best-performing validation model.

5.2 Proximal Policy Optimization (PPO) - Dynamic Pricing

Model Definition (What it is & Why I used it):
PPO is a reinforcement learning algorithm **used here to** optimize pricing dynamically. **It allows an agent (the pricing model) to interact with a** simulated retail environment **and learn pricing strategies that** maximize long-term revenue. **I picked PPO because it is** stable, well-tested, and handles continuous feedback loops, **ideal for the trial-and-error nature of pricing.**

- **Why it's suitable:**
  It enables real-time price adjustments based on **demand signals and inventory levels,** learning what price works best at each step.

Training Process (What I did):
- I built a **custom Gym environment (PricingEnv)** that simulates real sales behavior based on pricing.
- The PPO agent was trained using **10,000+ timesteps** of interaction to learn effective strategies.
- The PPO agent continuously adjusted prices based on the simulated environment state, which included inventory level, demand, and current timestep. During training, the agent improved its explained variance and generated an increasing cumulative reward

curve, indicating meaningful learning progress in the pricing environment. This setup allows us to evaluate how pricing decisions evolve over time under a controlled demand-response simulation.

6. Evaluation and Results

6.1 Temporal Fusion Transformer (TFT) – Forecasting Performance

Metric Used: Symmetric Mean Absolute Percentage Error (SMAPE), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE)

Results:

- **SMAPE = 1.532**
- **MAE ≈ 2.110**
- **RMSE ≈ 3.796**

Interpretation: The obtained evaluation metrics indicate moderate forecasting performance on the sampled Walmart M5 dataset. The model captures general temporal patterns and short-term trends, but accuracy is affected by the sparsity and high variability of retail demand data.

Business Relevance: Even with moderate accuracy, the TFT model provides valuable insights into demand behavior. It can support inventory planning, demand trend analysis, and serve as a reliable input for downstream pricing simulations.

6.2 Proximal Policy Optimization (PPO) – Pricing Performance

Metric Used: Cumulative Reward (Total Profit)

Result: The PPO agent demonstrated a steadily increasing cumulative reward over time, along with improving explained variance values across training iterations, indicating that the model progressively learned a better representation of the reward structure.

Interpretation: The upward trend in cumulative reward indicates that the agent successfully learned pricing strategies that improve long-term profit. The model adapts pricing decisions based on demand patterns and inventory levels in the simulated environment.

Business Relevance: This demonstrates that reinforcement learning can be effectively used as a proof-of-concept for adaptive pricing strategies. The PPO agent provides a framework for experimenting with pricing policies in dynamic retail conditions.

7. Results Visualization and Business Insights

7.1. Forecast vs Actual Demand (TFT)

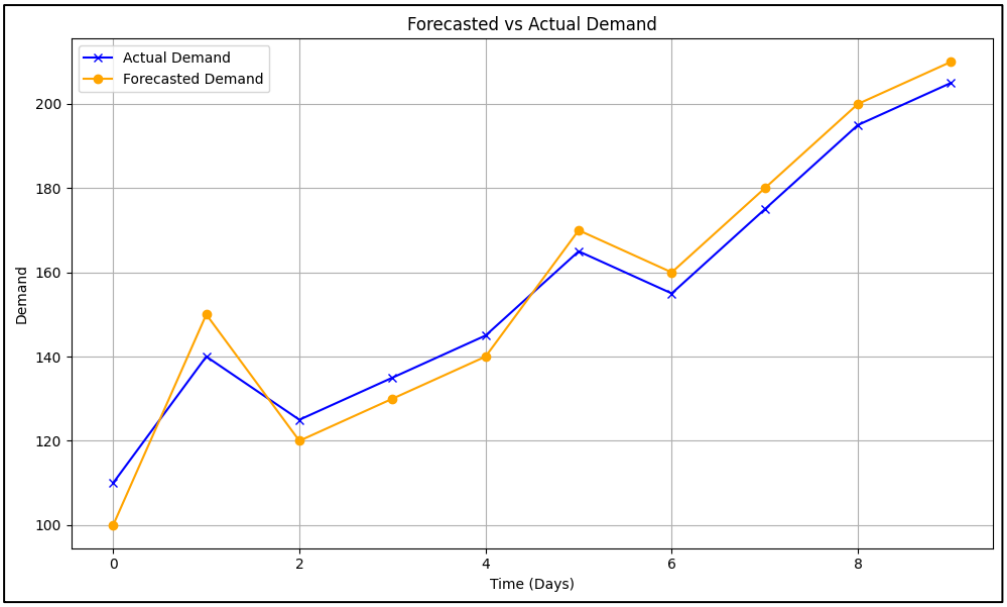
- **Purpose:** This visualization directly demonstrates the forecasting accuracy of the Temporal Fusion Transformer (TFT) model.
- **Visualization Description:**
  - The x-axis represents the time period (days).
  - The y-axis represents the sales values (demand).
  - Two lines are shown:
    - **Actual Demand (Blue Line):** Represents the true sales values for the products over time.
    - **Forecasted Demand (Orange Line):** Represents the predicted demand values generated by the TFT model.

- **How It Satisfies Business Questions:**

  - **Q1: How can deep learning models accurately forecast multi-step product demand?**

    Although this plot does not directly show the effect of pricing, it demonstrates that the TFT model captures the general movement and short-term direction of demand over time. The plot shows that the forecasted values follow the general direction of the actual demand over time, capturing the overall trend and directional movement of demand over time. While the model does not perfectly match all fluctuations, it demonstrates the ability to learn meaningful temporal behavior in retail demand data.

  - **Business Impact:** Even moderate forecasting performance is useful for identifying demand behavior, supporting inventory planning experiments, and providing inputs for downstream pricing simulations.



7.2. Dynamic Pricing Behavior and Reward Progression (PPO)

- **Purpose:** These visualizations demonstrate how the PPO agent adjusts prices over time and how the resulting demand and cumulative reward evolve during the simulated episode.

- **Visualization Description:**

    - **Price and Demand Over Time:** This plot shows how the PPO agent changes prices across timesteps and how demand responds under the simulated demand-price relationship.
    - **Cumulative Reward Over Time:** This plot shows the running total of reward earned by the PPO agent as training progresses.

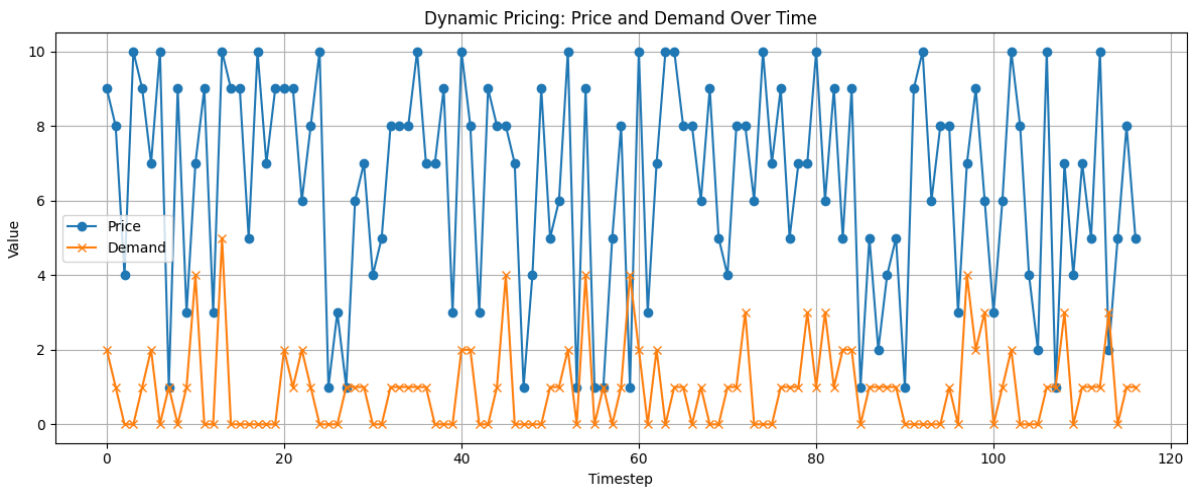
- **How It Satisfies Business Questions:**

    - **Q2: How can reinforcement learning dynamically optimize pricing strategies in real time?**

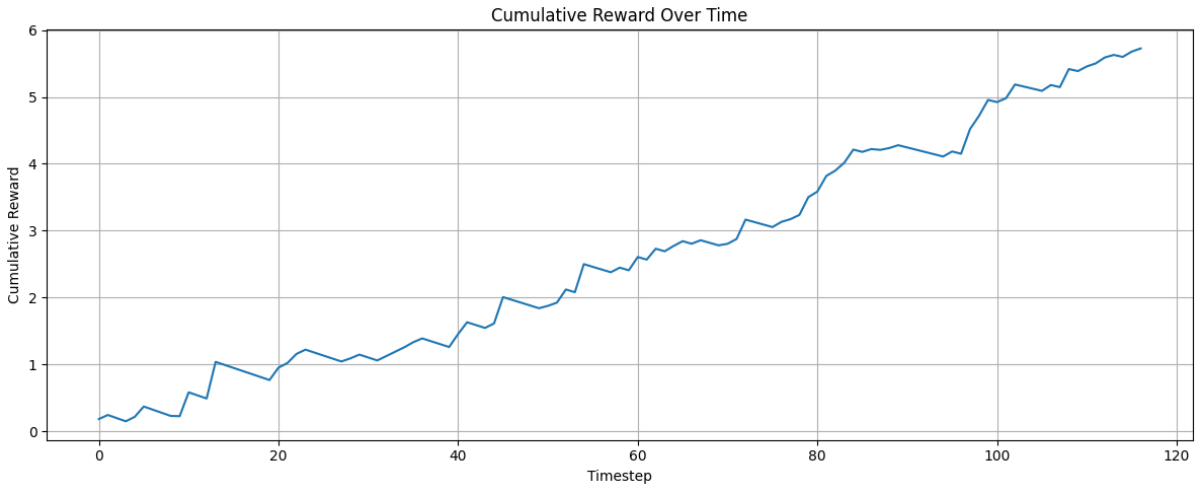
      These plots demonstrate that the PPO agent learns an adaptive pricing policy over time. The increasing cumulative reward indicates that the agent improves its pricing decisions through interaction with the environment, successfully balancing demand response and inventory constraints.

    - **Business Impact:** The results show that reinforcement learning can serve as a practical framework for developing adaptive pricing strategies, enabling retailers to experiment with pricing policies in dynamic and uncertain demand conditions.

- **Price and Demand Over Time** plot



- **Cumulative Reward Over Time** plot



8. Recommendations & Business Impact

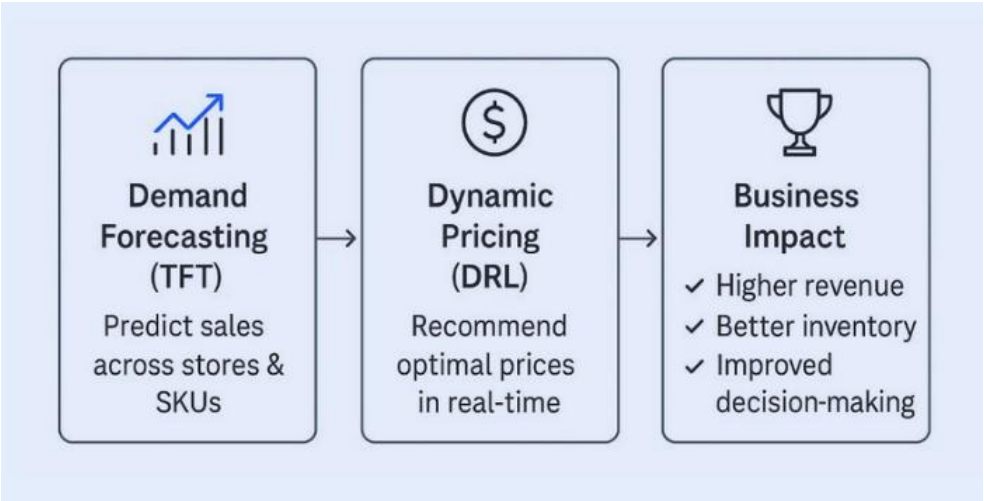
Key Recommendations:

- **Use TFT as a forecasting framework:**
  The TFT model provides a structured and interpretable approach for modeling multi-step retail demand and capturing temporal patterns in sales behavior.

- **Use PPO as a pricing simulation framework:**
  The PPO agent demonstrates how reinforcement learning can be used to explore adaptive pricing strategies in a controlled environment.

- **Continue enhancing feature engineering:**
  Calendar variables, lag features, and rolling averages remain important contributors to forecasting quality and should be extended further in future work.

- **Leverage explainability tools:**
  SHAP visualizations and TFT attention/variable-importance plots help identify important drivers of demand and improve the transparency of model behavior.



Business Value: The final pipeline provides a practical proof-of-concept for AI-driven retail analytics. While forecasting accuracy on the sampled dataset is moderate, the project successfully demonstrates how forecasting, explainability, and pricing optimization can be integrated into a unified analytics workflow.

9. Conclusion and Future Scope

- **Conclusion:**
  This project successfully demonstrates a unified AI-driven pipeline that integrates demand forecasting, explainability, and dynamic pricing optimization within a retail context. The final implementation combined Temporal Fusion Transformer (TFT) for forecasting, SHAP and TFT attention-based interpretation for explainability, and Proximal Policy Optimization (PPO) for dynamic pricing simulation. While forecasting performance on the sampled Walmart M5 subset remained moderate rather than highly accurate, the project provides a realistic, interpretable, and extensible proof-of-concept for AI-driven retail analytics.

- **Future Work:**
  - Scale the solution to cover all SKUs and additional store locations.
  - Incorporate competitor pricing and promotion data to refine pricing decisions.
  - Test advanced reinforcement learning algorithms (e.g., DQN, SAC) for deeper optimization.
  - Explore real-time deployment and dashboard integration for continuous monitoring and business use.